



Seep Pahuja
Biology Educator
(NEET UG)



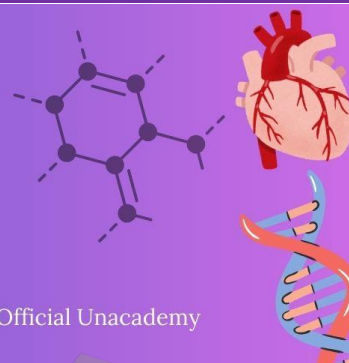
seep.pahuja



Seep Pahuja Biology - Official Unacademy

LAST 10 YEARS

NEET PYQ



Chapter 10 : Biotechnology and its Applications

1. Which of the following genetically engineered organisms was used by Eli Lilly to prepare human insulin? [NEET-2025]

- | | |
|---------------|-----------|
| (1) Bacterium | (2) Yeast |
| (3) Virus | (4) Phage |

2. Match List-I with List-II :

[Re-NEET-2024]

List-I

- A. Genetically engineered Human Insulin
- B. GM Cotton
- C. ADA Deficiency
- D. ELISA

List-II

- I. Gene therapy
- II. E. coli
- III. Antigen-antibody interaction
- IV. Bacillus thuringiensis

Choose the correct answer from the options given below:

- | | |
|----------------------------|----------------------------|
| (1) A-III, B-II, C-IV, D-I | (2) A-II, B-I, C-IV, D-III |
| (3) A-IV, B-III, C-I, D-II | (4) A-II, B-IV, C-I, D-III |

3. Given The Bt toxin in genetically engineered Bt cotton kills the pest by:

[Re-NEET-2024]

- | | |
|-------------------------------------|-------------------------------------|
| (1) Creating pores in the midgut | (2) Damaging the respiratory system |
| (3) Degenerating the nervous system | (4) Altering the pH of body fluids |

4. Given below are two statements :

[Re-NEET-2024]

Statement I : RNA interference takes place in all Eukaryotic organisms as method of cellular defense.

Statement II : RNAi involves the silencing of a specific mRNA due to a complementary single-stranded RNA molecule that binds and prevents translation of mRNA

In the light of the above statements, choose the correct answer from the options given below.

- (1) Statement I is true but Statement II is false.
- (2) Statement I is false but Statement II is true.
- (3) Both Statement I and Statement II are true.
- (4) Both Statement I and Statement II are false.

5. Given below are two statements: [NEET-2024]

Statement I : Bt toxins are insect group specific and coded by a gene cry IAc.

Statement II : Bt toxin exists as inactive protoxin in *B. thuringiensis*. However, after ingestion by the insect the inactive protoxin gets converted into active form due to acidic pH of the insect gut.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both Statement I and Statement II are true
- (2) Both Statement I and Statement II are false
- (3) Statement I is true but Statement II is false
- (4) Statement I is false but Statement II is true

6. Match List I with List II : [NEET-2024]

List I

List II

A. α -I antitrypsin

I. Cotton bollworm

B. Cry Iab

II. ADA deficiency

C. Cry IAc

III. Emphysema

D Enzyme replacement therapy

IV. Corn borer

Choose the correct answer form the options given below:

- (1) A-II, B-I, C-IV, D-III
- (2) A-III, B-I, C-II, D-IV
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-IV, C-I, D-III

7. Select incorrect statement, regarding chemical structure of insulin. [NEET-2023-MANIPUR]

- (1) C-peptide is not present in mature insulin molecule.
- (2) Polypeptide chains A and B are linked by disulphide bridges.
- (3) Mature insulin molecule consists of three polypeptide chains - A, B and C.
- (4) Insulin is synthesized as prohormone which contains extra stretch of C-peptide.

8. Match List - I with List - II. [NEET-2023-MANIPUR]

List - I

List - II

(A) Gene therapy

(I) Separation of DNA fragments

(B) RNA interference

(II) Diagnostic test for AIDS

(C) ELISA

(III) Cellular defence

(D) Gel Electrophoresis

(IV) Allows correction of a gene defect

Choose the correct answer from the options given below :

- (1) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
- (2) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)
- (3) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
- (4) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

9. Which one of the following techniques does not serve the purpose of early diagnosis of a disease for its early treatment? [NEET-2023]

- (1) Recombinant DNA Technology
- (2) Serum and Urine analysis
- (3) Polymerase Chain Reaction (PCR) technique
- (4) Enzyme Linked Immuno-Sorbent Assay (ELISA) technique

10. Transposons can be used during which one of the following? [NEET-2022]

- (1) Gene silencing
- (2) Autoradiography
- (3) Gene sequencing
- (4) Polymerase Chain Reaction

11. Statements related to human Insulin are given below :- Which statement(s) is/ are correct about genetically engineered Insulin ? [NEET-2022]

- (a) Pro-hormone insulin contain extra stretch of C-peptide
- (b) A-peptide and B-peptide chains of insulin were produced separately in E.coli, extracted and combined by creating disulphide bond between them.
- (c) Insulin used for treating Diabetes was extracted from Cattles and Pigs.
- (d) Pro-hormone Insulin needs to be processed for converting into a mature and functional hormone.
- (e) Some patients develop allergic reactions to the foreign insulin.

Choose the most appropriate answer from the options given below:-

- (1) (b)only
- (2) (c) and (d) only
- (3) (c), (d) and (e) only
- (4) (a), (b) and (d) only

12. Milk of transgenic 'Cow Rosie' was nutritionally more balanced product for human babies than natural cow milk because it contained :- [Re-NEET-2022]

- (1) Human protein α -1-antitrypsin
- (2) Human alpha-lactalbumin
- (3) Human insulin-like growth factor
- (4) Human enzyme Adenosine Deaminase (ADA)

13. For effective treatment of the disease, early diagnosis and understanding its pathophysiology is very important. Which of the following molecular diagnostic techniques is very useful for early detection? [NEET-2021]

- (1) Western Blotting Technique
- (2) Southern Blotting Technique
- (3) ELISA Technique
- (4) Hybridization Technique

14. With regard to insulin choose correct options. [NEET-2021]

- (a) C-peptide is not present in mature insulin.
- (b) The insulin produced by rDNA technology has C-peptide.
- (c) The pro-insulin has C-peptide.
- (d) A-peptide and B-peptide of insulin are interconnected by disulphide bridges.

Choose the correct answer from the options given below.

- (1) (b) and (d) only
- (2) (b) and (c) only
- (3) (a), (c) and (d) only
- (4) (a) and (d) only

15. When gene targetting involving gene amplification is attempted in an individual's tissue to treat disease, it is known as :- [NEET-2021]

- (1) Biopiracy
- (2) Gene therapy
- (3) Molecular diagnosis
- (4) Safety testing

16. Bt cotton variety that was developed by the introduction of toxin gene of *Bacillus thuringiensis* (Bt) is resistant to :- [NEET-2020]

- (1) Insect predators
- (2) Insect pests
- (3) Fungal diseases
- (4) Plant nematodes

17. Match the following columns and select the correct option:- [NEET-2020]

Column - I

Column - II

- | | |
|---------------|--|
| (a) Bt cotton | (i) Gene therapy |
| (b) Adenosine | (ii) Cellular defence deaminase deficiency |
| (c) RNAi | (iii) Detection of HIV infection |
| (d) PCR | (iv) <i>Bacillus thuringiensis</i> |

Choose the correct answer from the options given below.

- (a) (b) (c) (d)
- (1) (i) (ii) (iii) (iv)
- (2) (iv) (i) (ii) (iii)
- (3) (iii) (ii) (i) (iv)
- (4) (ii) (iii) (iv) (i)

18. Which of the following statements is not correct? [NEET-2020]

- (1) Genetically engineered insulin is produced in *E. coli*.
- (2) In man insulin is synthesised as a proinsulin.
- (3) The proinsulin has an extra peptide called C-peptide.
- (4) The functional insulin has A and B chains linked together by hydrogen bonds.

19. RNA interference is used for which of the following purposes in the field of biotechnology ?
(1) to develop a plant tolerant to abiotic stresses [NEET-2020(COVID-19)]
(2) to develop a pest resistant plant against infestation by nematode
(3) to enhance the mineral usage by the plant
(4) to reduce post-harvest losses
20. The laws and rules to prevent unauthorised exploitation of bio-resources are termed as-
(1) Biopatenting (2) Bioethics [NEET-2020(COVID-19)]
(3) Bioengineering (4) Biopiracy
21. Which of the following is true for Golden rice? [NEET-2019]
(1) It is Vitamin A enriched, with a gene from daffodil
(2) It is pest resistant, with a gene from *Bacillus thuringiensis*
(3) It is drought tolerant, developed using *Agrobacterium* vector
(4) It has yellow grains, because of a gene introduced from a primitive variety of rice
22. Which of the following features of genetic code does allow bacteria to produce human insulin by recombinant DNA technology? [NEET-2019]
(1) Genetic code is not ambiguous (2) Genetic code is redundant
(3) Genetic code is nearly universal (4) Genetic code is specific
23. What triggers activation of protoxin to active Bt toxin of *Bacillus thuringiensis* in boll worm?
(1) Body temperature (2) Moist surface of midgut [NEET-2019]
(3) Alkaline pH of gut (4) Acidic pH of stomach
24. Exploitation of bioresources of a nation by multinational companies without authorization from the concerned country is referred to as: - [NEET-2019(ODISHA)]
(1) Bioweapon (2) Biopiracy
(3) Bioethics (4) Biowar
25. In RNAi, the genes are silenced using:- [NEET-2019(ODISHA)]
(1) ds-RNA (2) ss-DNA
(3) ss-RNA (4) ds-DNA
26. In India, the organisation responsible for assessing the safety of introducing genetically modified organisms for public use is :- [NEET-2018]
(1) Indian Council of Medical Research (ICMR)
(2) Council for Scientific and Industrial Research (CSIR)
(3) Research Committee on Genetic Manipulation (RCGM)
(4) Genetic Engineering Appraisal Committee (GEAC)

27. A 'new variety of rice was patented by a foreign company though such varieties have been present in India for a long time. This is related to :- [NEET-2018]

- | | |
|----------------|---------------------|
| (1) Co-667 | (2) Sharbati Sonora |
| (3) Lerma Rojo | (4) Basmati |

28. Use of bioresources by multinational companies and organisations without authorisation from the concerned country and its people is called :- [NEET-2018]

- | | |
|----------------------|---------------------|
| (1) Bio-infringement | (2) Biopiracy |
| (3) Biodegradation | (4) Bioexploitation |

29. The two polypeptides of human insulin are linked together by :- [NEET- I 2016]

- | | |
|--------------------|-------------------------|
| (1) Hydrogen bonds | (2) Phosphodiester bond |
| (3) Covalent bond | (4) Disulphide bridges |

30. Which kind of therapy was given in 1990 to a four year old girl with adenosine deaminase (ADA) deficiency ? [NEET- II 2016]

- | | |
|-------------------|-----------------------|
| (1) Immunotherapy | (2) Radiation therapy |
| (3) Gene therapy | (4) Chemotherapy |

ANSWER KEY

1. (1)
2. (4)
3. (1)
4. (1)
5. (3)
6. (3)
7. (3)
8. (1)
9. (2)
10. (1)
11. (1)
12. (2)
13. (3)
14. (3)
15. (2)
16. (2)
17. (2)
18. (4)
19. (2)
20. (1)
21. (1)
22. (3)
23. (3)
24. (2)
25. (1)
26. (4)
27. (4)
28. (2)
29. (4)
30. (3)